

AI-Based Crowd & River Health Monitoring System

Smart Monitoring for Sustainable Religious Events



Crowd Monitoring



River Health Monitoring



Real-time Alerts



Smart Decision Support



TEAM MEMBERS



Khushi Kushwah



Daksh Khandelwal



॥ सा विद्या या विमुक्तये ॥

IIT JODHPUR

भारतीय प्रौद्योगिकी संस्थान जोधपुर

Indian Institute of Technology Jodhpur

Problem Statement

Mass religious gatherings put immense stress on both people and the river ecosystem.



Crowd Monitoring Challenges



Overcrowding during Purnima, Amavasya, Sankranti, Kumbh, etc.



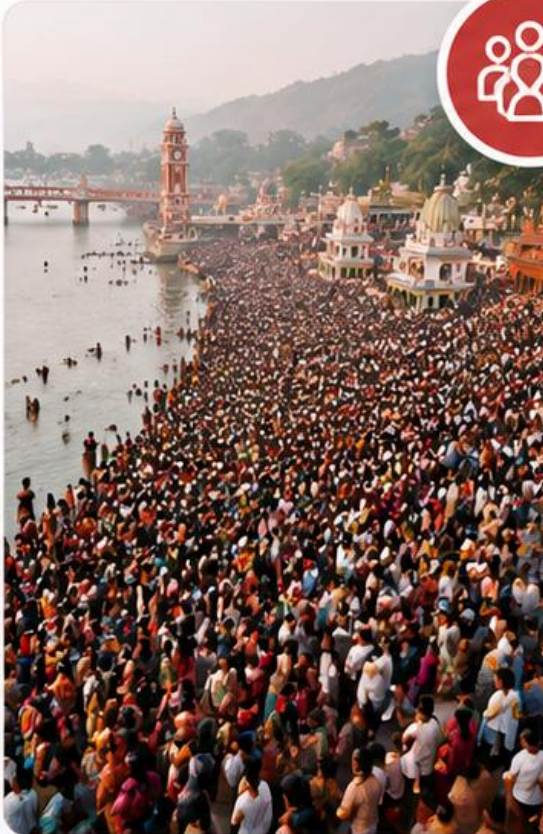
Stampede risk due to uncontrolled crowd movement



Delayed emergency response



Lack of real-time crowd analytics



River Health Challenges



Sudden deterioration in water quality



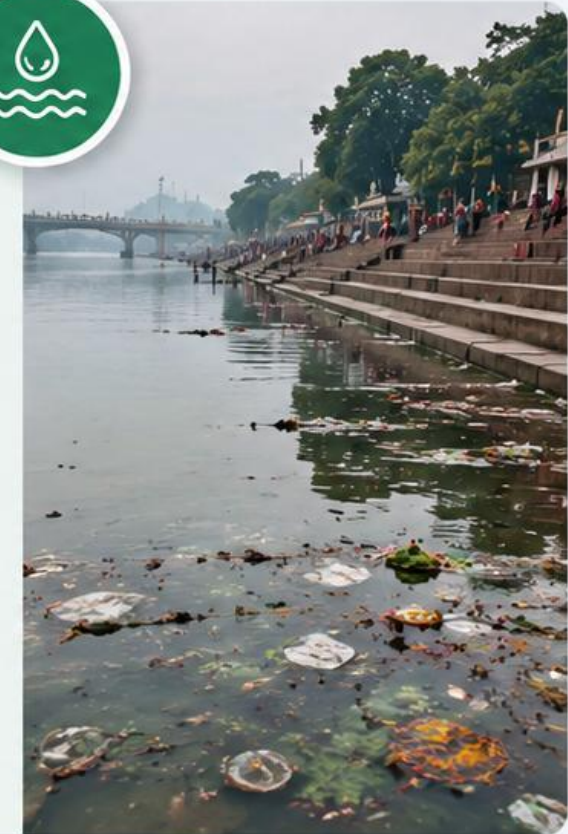
Increased turbidity and organic pollution



Riverbank erosion due to heavy footfall



Ecological stress from repeated mass gatherings



These challenges demand an integrated, AI-powered solution for real-time monitoring and smart decision-making.



Why Current Systems Are Not Enough

Existing approaches are reactive, fragmented and lack real-time intelligence.



Manual Crowd Monitoring

Labour-intensive and highly error-prone.



CCTV with Limited Coverage

Blind spots and limited situational awareness.



Periodic Water Sampling

Not continuous, misses sudden deterioration.



Separate Monitoring of Crowd & Environment

No integration between people and ecosystem data.



No Predictive Analysis

Lack of forecasting leads to unprepared responses.



Delayed Decision-Making

Critical actions taken too late.

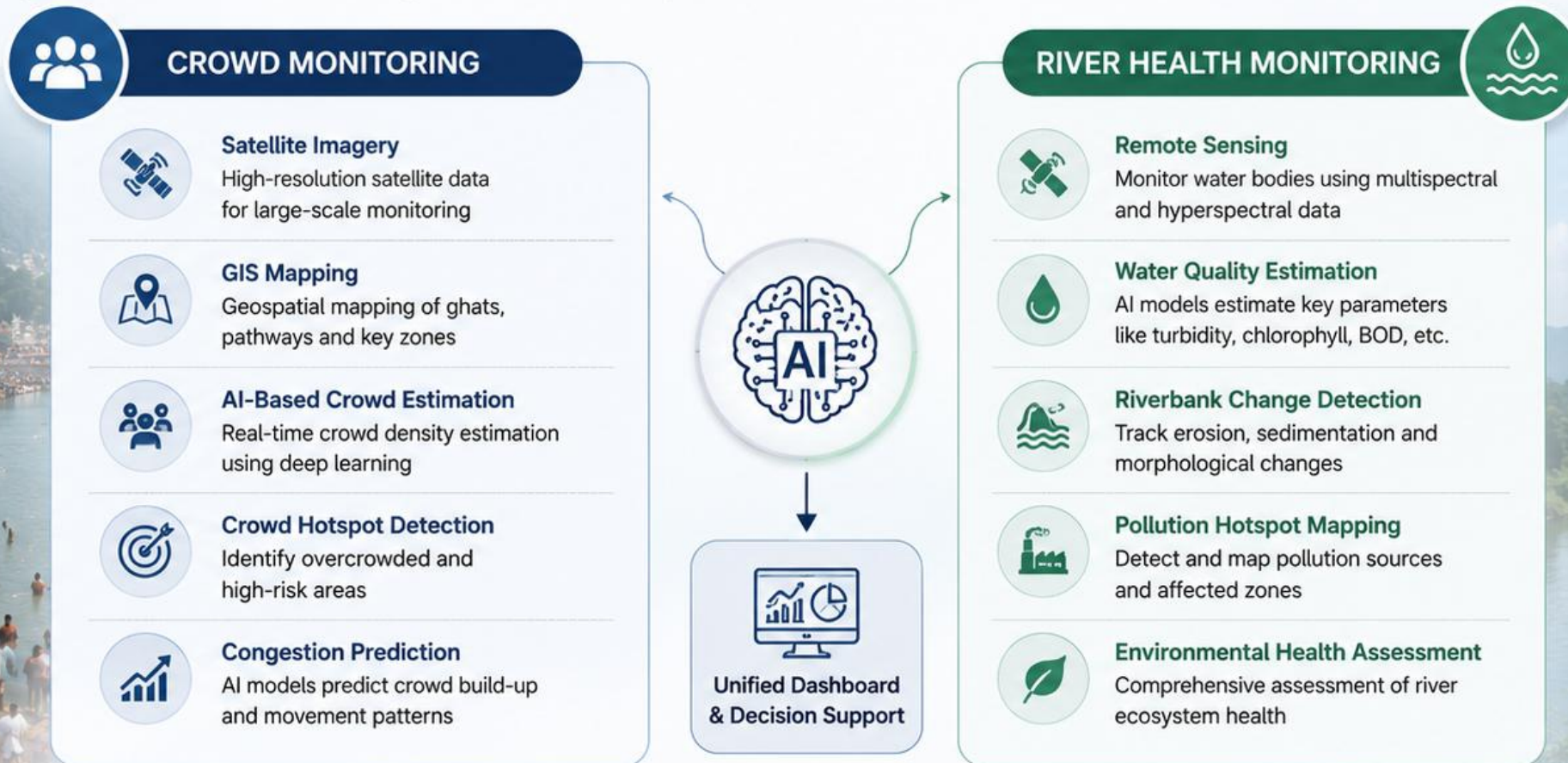


We need an integrated, AI-powered system for real-time monitoring and early warning.

Proposed AI-Based Solution

Integrated Smart Monitoring Platform

Combining advanced AI, remote sensing and real-time analytics for safer events and healthier rivers.



One Integrated Platform. Real-time Insights. Smarter Decisions.

AI SOLUTION ARCHITECTURE

From Multi-Source Data to Smart Decisions

A complete AI pipeline for real-time crowd and river health monitoring.

INPUT



Satellite Images

High-resolution multispectral imagery



GIS Data

Ghats, zones, infrastructure, river network



Weather Data

Rainfall, temperature, humidity, wind



Historical Event Data

Past crowd, water quality, and event records

AI PROCESSING



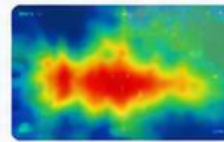
Crowd Detection

AI models detect people from satellite/CCTV



Density Estimation

Real-time crowd density calculation



Water Quality Analysis

AI models analyze water quality parameters



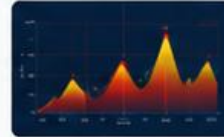
Riverbank Monitoring

Detect erosion, sedimentation and morphological changes



Risk Prediction

Predict crowd surge, pollution and environmental risks

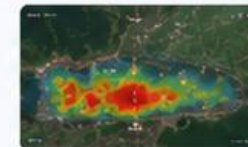


OUTPUT



Crowd Risk Maps

Live density heatmaps and risk zones



River Health Maps

Water quality and environmental status



Environmental Alerts

Real-time warnings and notifications



Decision Support Dashboard

Unified dashboard for authorities



Integrated AI. Real-time Insights. Smarter Actions.

RIVER HEALTH MONITORING SOLUTION

AI continuously monitors key environmental parameters for a healthier river ecosystem.

AI MONITORS



WATER QUALITY INDEX

AI computes WQI using multiple water quality parameters.



TURBIDITY

Monitors water clarity and suspended particles.



CHLOROPHYLL CONCENTRATION

Detects algal blooms and aquatic productivity levels.



ORGANIC MATTER

Estimates organic load and potential pollution sources.



RIVERBANK STABILITY

AI detects erosion, land changes and bank weakening.



POLLUTION HOTSPOTS

Identifies areas with high pollution and contamination risk.



AI PREDICTS



HIGH POLLUTION ZONES

Predicts areas likely to exceed safe pollution levels.



ENVIRONMENTAL DEGRADATION

Forecasts long-term environmental stress and degradation.



WATER QUALITY CHANGES

Predicts changes in water quality before and after events.



TREND ANALYSIS

Identifies patterns and trends for future planning.



EARLY ALERTS

Sends real-time alerts for quick response.



AI ensures continuous, accurate and proactive monitoring for long-term river health and ecological balance.



CROWD MONITORING SOLUTION

AI continuously monitors, analyzes and predicts crowd behavior for safer management.

AI DETECTS



CROWD DENSITY

Real-time estimation of people density in different zones.



CROWD MOVEMENT

Tracks movement patterns and flow directions.



HIGH-RISK ZONES

Identifies areas with extreme crowd concentration.



ENTRY-EXIT CONGESTION

Monitors choke points and congestion at access routes.

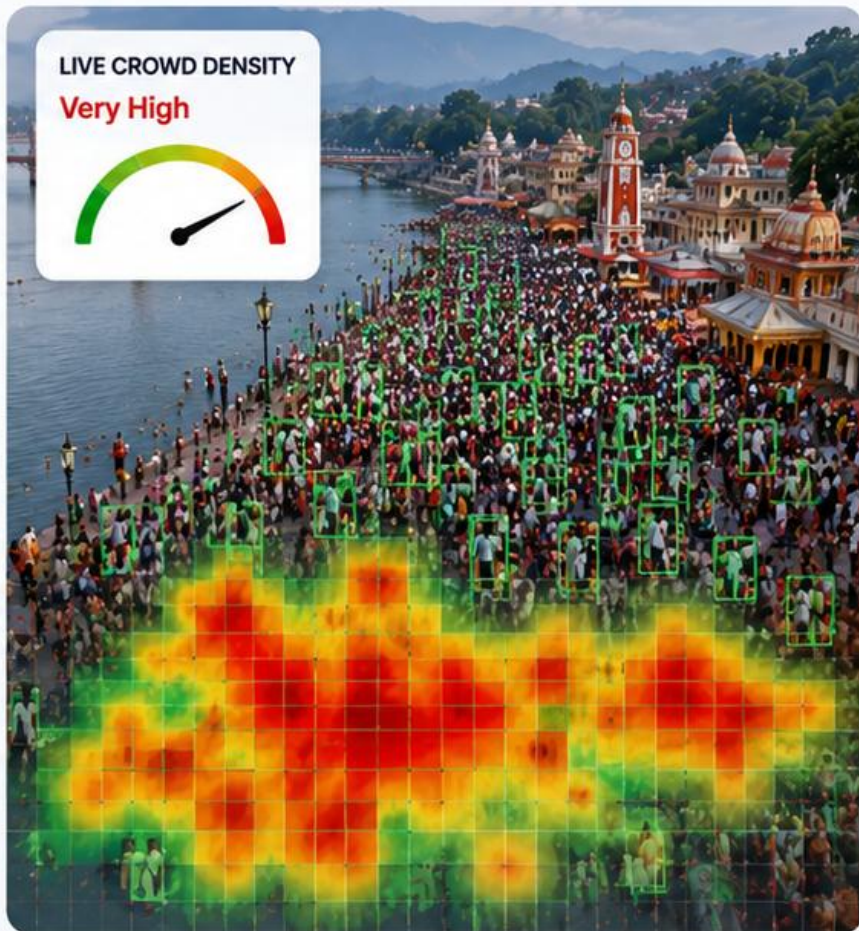


BEHAVIOR PATTERNS

Analyzes crowd behavior for anomaly detection.

LIVE CROWD DENSITY

Very High



AI PREDICTS



CROWD SURGE

Predicts sudden increase in crowd in specific areas.



BOTTLENECKS

Detects potential bottlenecks before they occur.



SAFE CROWD CAPACITY

Estimates safe capacity for each zone.



EMERGENCY EVACUATION ROUTES

Suggests optimal evacuation paths in real-time.



RISK ALERTS

Sends real-time alerts to authorities for action.



HIGH CROWD RISK

Har Ki Pauri Ghat

Density: 92%

Take immediate action



AI-Powered Insights for Safer Crowds and Smarter Decisions.



SMART DECISION SUPPORT DASHBOARD

Real-time insights for informed decisions and proactive management.



LIVE CROWD DENSITY MAP

Real-time crowd distribution across all zones.



RIVER HEALTH STATUS

Live monitoring of water quality and river conditions.



POLLUTION RISK ALERTS

Instant alerts for high pollution and risk zones.



SAFE BATHING ZONES

AI-recommended safe zones for public use.



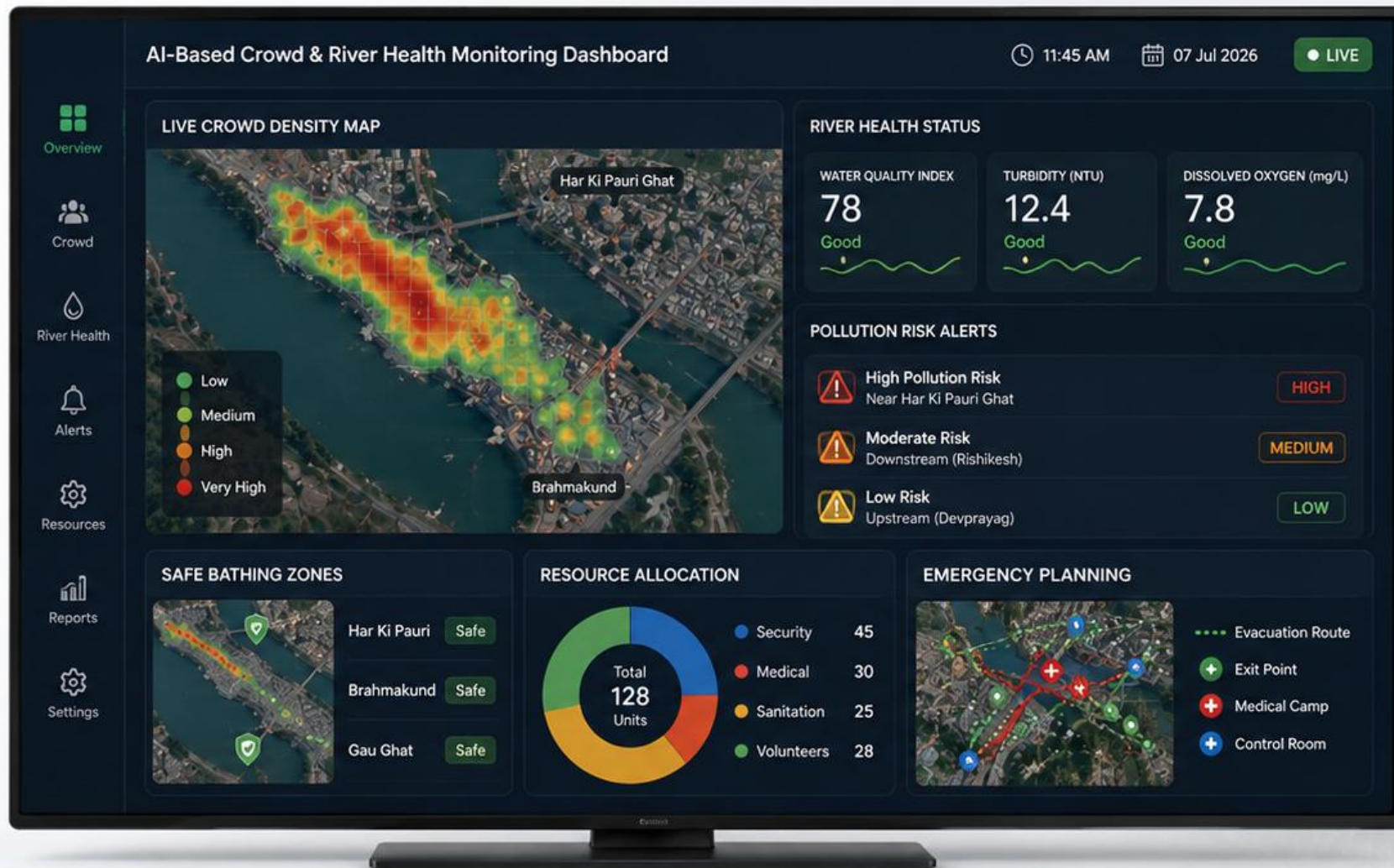
RESOURCE ALLOCATION

Smart allocation of staff, medical camps, and facilities.



EMERGENCY PLANNING

Evacuation routes and real-time emergency support.



One Dashboard. Real-time Data. Smarter Decisions.

EXPECTED OUTCOMES

Better management for people, environment and future.



FOR ADMINISTRATION



BETTER CROWD MANAGEMENT

Real-time insights ensure smooth crowd flow and reduced congestion.



FASTER EMERGENCY RESPONSE

AI-powered alerts enable quick action in critical situations.



DATA-DRIVEN PLANNING

Historical analytics and predictions improve future event planning.



EFFICIENT RESOURCE ALLOCATION

Optimize deployment of staff, infrastructure and services.



ENHANCED PUBLIC SAFETY

Reduced risks, better monitoring and informed decision-making.



AI for People Safety.
AI for a Healthier River.



FOR ENVIRONMENT



CLEANER RIVERS

Continuous monitoring helps reduce pollution and maintain water quality.



REDUCED POLLUTION

Early detection of pollution hotspots enables proactive action.



PROTECTION OF RIVERBANKS

AI-based erosion monitoring helps prevent degradation.



ECOLOGICAL BALANCE

Supports biodiversity and maintains ecosystem health.



SUSTAINABLE RELIGIOUS TOURISM

Balanced approach ensures long-term benefits for people and nature.



SAFER PEOPLE



HEALTHIER RIVERS



SMARTER PLANNING



SUSTAINABLE FUTURE



Integrated Insights. Responsible Actions. Lasting Impact.

TECHNOLOGIES & TOOLS

Modern technologies powering real-time insights and smarter decisions.

TECHNOLOGIES



ARTIFICIAL INTELLIGENCE / ML
Machine learning models for detection, prediction and anomaly analysis.



REMOTE SENSING
Satellite imagery and multispectral data for large-scale monitoring.



GIS & SPATIAL ANALYTICS
Geospatial analysis for mapping, visualization and spatial insights.



CLOUD COMPUTING
Scalable, secure and flexible infrastructure for real-time data processing.



BIG DATA PROCESSING
Efficient handling of large volumes of heterogeneous data.



DATA VISUALIZATION
Interactive dashboards for clear, actionable insights.



TOOLS & PLATFORMS



Google Earth Engine
Satellite data access and analysis.



QGIS
Open-source GIS for mapping and spatial analysis.



Python
Core programming for AI models, analysis and automation.



TensorFlow / Scikit-learn
Machine learning model development and deployment.



PostgreSQL / PostGIS
Spatial database for efficient data management.



Power BI / Tableau
Interactive dashboards and visual analytics.



The Right **Technology**. The Right **Insights**. The Right **Impact**.



Thank You

— Questions & Discussion —



Safer Rivers
Stronger Communities



Sustainable
Ecosystems



Smarter Decisions
Better Outcomes



Integrated Systems
Stronger Governance

Khushi Kushwah

Daksh Khandelwal



Indian Institute of Technology (IIT) Jodhpur